

# MUHAMMED ARSHAQ

Full-Stack Software Engineer

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## PROFESSIONAL SUMMARY

Full-stack software engineer with 5+ years of experience designing, building and shipping web, mobile and AI-enabled products for startups and SMEs across Sri Lanka, the GCC and international markets. Experienced across the full product lifecycle, from requirements and system architecture to real-time features, payment integration, IoT telemetry and computer vision, with a consistent record of taking products from concept to production. As a co-founder of a software agency, brings strong ownership, client communication and delivery discipline. BSc (Hons) in Software Engineering from SLIIT (Second Class Upper, W GPA 3.63) with applied computer-vision research and an IEEE conference publication.

## TECHNICAL SKILLS

**Languages:** TypeScript, JavaScript, Java, Python, PHP, SQL, C, C++, HTML, CSS

**Frameworks & Libraries:** React, Next.js, React Native, Node.js, NestJS, Express.js, Redux, Zustand, Tailwind CSS

**Architecture:** Microservices, REST, gRPC, WebSockets (Socket.io), event-driven systems, API gateways

**AI / ML:** PyTorch, Ultralytics YOLOv8, ONNX Runtime, computer vision, model training & deployment

**Databases & ORM:** PostgreSQL, MongoDB, MySQL, SQLite, Prisma ORM

**DevOps & Testing:** Docker, Docker Compose, NX monorepo, Git/GitHub, Linux, Jest, Playwright, Grafana k6

**Spoken Languages:** English (professional), Sinhala, Tamil

## PROFESSIONAL EXPERIENCE

### FocalDive (Pvt) Ltd - Full-Stack Software Engineer & Co-Founder

Feb 2021 - Present

Software agency: web, mobile, booking, e-commerce, SaaS and AI solutions

Puttalam, Sri Lanka

- Design, build and ship full-stack web, mobile and AI products for startups and SMEs in Sri Lanka, the GCC and global markets, from architecture through deployment.
- Delivered production platforms with real-time tracking, payments and wallets, IoT fleet telemetry, booking engines and admin dashboards (see Selected Projects).
- Design backend services and APIs, including microservice and event-driven architectures, with a focus on reliability, maintainability and performance under load.
- Lead technical delivery in Agile teams, covering requirements, estimation, code reviews and releases across 15+ client projects, and mentor junior developers and interns.

### IMARA Software Solutions

Nov 2021 - Nov 2024

Software development company building custom web and mobile applications for business clients

Puttalam, Sri Lanka

#### Web Application Developer (Full-time)

Jul 2022 - Nov 2024

- Developed and maintained production MERN-stack web applications, including CRM functionality, focusing on performance, maintainability and user experience.
- Integrated REST APIs to fetch and manipulate data, streamlining communication between frontend clients and backend services.

#### Mobile Application Developer - Intern (Full-time)

Nov 2021 - Jun 2022

- Built cross-platform mobile app features, translating Figma designs into production screens and integrating backend APIs.

## RESEARCH

### BrineX - Integrated Climate-Intelligent Salt Production Ecosystem

2025 - 2026

Final-Year Research Project, SLIIT (Ref: 25-26J-431) - four-member team

Puttalam, Sri Lanka

- Four-module AI platform (forecasting, planning, vision quality control, federated learning) built as NestJS microservices in an NX monorepo.

#### My component: Salt Crystal Impurity Detection and Quality Grading Using Real-Time Computer Vision

- Designed and delivered the platform's computer-vision quality-control module, the first AI-based visual inspection system for tropical salt production, detecting pure, impure and foreign-material classes.
- Built a custom dataset from field capture at Puttalam salterns: 650 annotated images (Label Studio, CVAT) expanded to 1,952 via augmentation; benchmarked MobileNetV2, ResNet-50 and YOLOv8 before selecting a detection approach.
- Achieved over 99% detection accuracy on unseen test images, correctly identifying 99.7% of impure salt samples - the class that matters most for product quality.

- Engineered a production NestJS/TypeScript microservice with ONNX Runtime inference, gRPC contracts, MongoDB persistence and Socket.io streaming to a Next.js 14 dashboard, holding under 50 ms per frame (5 FPS) on CPU-only hardware.

## PUBLICATIONS

T. Ansar, I. Sirimanna, **M. Arshaq**, R. Perumbuli, N. Silva and T. Jayalath, "From Pond to Market: A Two-Stage AI Forecasting System for Salt Production Planning in Puttalam, Sri Lanka," in *Proc. 8th Int. Congress on Human-Computer Interaction, Optimization and Robotic Applications (ICHORA 2026)*, IEEE Türkiye Section, Ankara, Türkiye, May 2026 (oral presentation).  
IEEE Xplore: [ieeexplore.ieee.org/document/11537206](https://ieeexplore.ieee.org/document/11537206)

## SELECTED PROJECTS

**Molt - Real-Time Livestock Monitoring System (FocalDive)** **Mar 2026 - Present**  
*DFCCNet density regression, IP cameras, SMS & in-app alerts*

- Replaced twice-daily manual poultry headcounts with continuous IP-camera monitoring using DFCCNet density regression, automated threshold alerts and SMS/in-app notifications. Standardised 600px frames and introduced a single inference queue to eliminate out-of-memory failures as camera count increased.
- Chose density regression over object detection to count overlapping birds accurately, and removed out-of-memory failures by standardising frames to 600 px behind a single inference queue so memory stays flat as cameras scale.

**PassQue - Patient Queue Management System (FocalDive)** **Apr 2026 - Present**  
*React Native, Spring Boot* *Role: Cross-Platform Mobile App Developer*

- Built the patient-facing React Native mobile app from scratch for a clinic queue system that digitises appointments and cuts physical waiting times.
- Delivered real-time queue position tracking, status notifications and an intuitive, accessible UI; integrated Spring Boot REST endpoints for queue, patient, doctor and clinic data.

**Weels - End-to-End Food Ordering & Delivery Platform (FocalDive)** **Oct 2025 - Jun 2026**  
*React Native, Nest.js, PostgreSQL*

- Two connected cross-platform apps (customer and delivery partner) on a shared backend: restaurant browsing, cart, secure checkout, live order tracking, rider assignment and run history.
- Pushed every rider state change to the customer app in real time and moved dispatch from phone calls into an in-system queue; PostgreSQL relational constraints prevent partial or duplicate order assignment.

**City Bikes & Segway - E-Scooter Rental & Tracking App, Qatar (FocalDive)** **Jan 2025 - Feb 2026**  
*Cross-platform mobile, IoT fleet telemetry, in-app payments*

- Self-service rental app for a Qatari micro-mobility operator covering live-map vehicle discovery, unlock, in-app payment and wallet-based 30-minute and 1-hour ride passes.
- Integrated IoT location telemetry into an admin dashboard for live fleet visibility, gated unlocks on payment authorisation, and built a bilingual English/Arabic UI with full RTL support.

**BMP - Sports Arena Booking & Management Platform (FocalDive)** **Apr 2025 - Nov 2025**  
*Next.js, full-stack*

- Unified players and turf owners on one calendar: location-based discovery with live availability and instant booking, plus an owner dashboard for slots, rates, discounts, payouts and month-on-month earnings.
- Prevented double bookings with a single source of truth and slot locking at confirmation; added recurring weekly slot configuration and a built-in wallet for balances, discounts and settlements.

## EDUCATION

**BSc (Hons) in Information Technology, Specialising in Software Engineering** **2022 - 2026**  
*Sri Lanka Institute of Information Technology (SLIIT)* *Malabe, Sri Lanka*

- Second Class Upper Division (WGPA 3.63 / 4.00); Dean's List x4; SLIIT Scholarship Award for academic performance

**G.C.E. Advanced Level - Physical Science (Physics, Chemistry, Combined Mathematics)** **2018 - 2020**  
*President's Science College* *Puttalam, Sri Lanka*